CONTROLLER

File: CONTROLLER [2].kicad_sch

GATE DRIVER

File: GATE DRIVER [3].kicad_sch

SENSORS

File: SENSORS.kicad_sch

POWER

File: POWER [1].kicad_sch

POWER MOSFET

File: POWER_MOSFET.kicad_sch

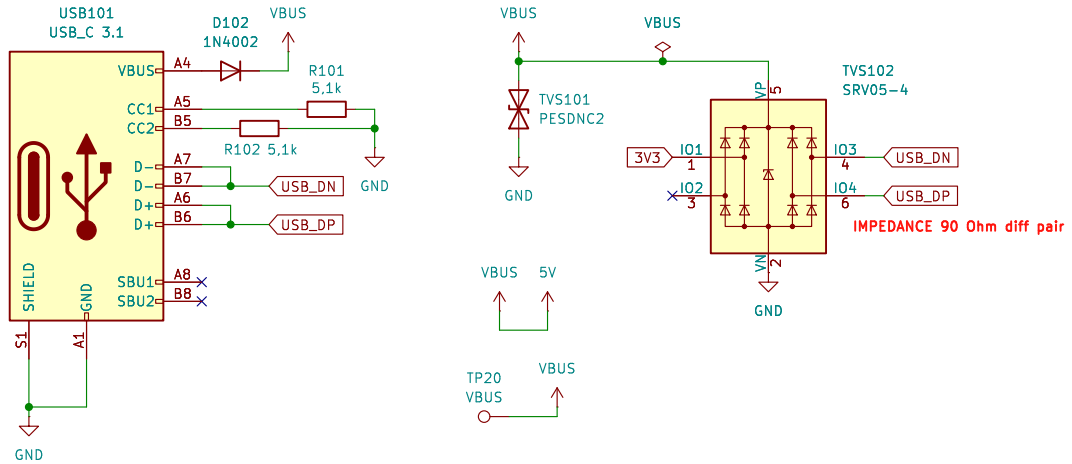
ANGEL PELEGRY VELASCO

Sheet: /
File: ESP32_Motor_Controller.kicad_sch

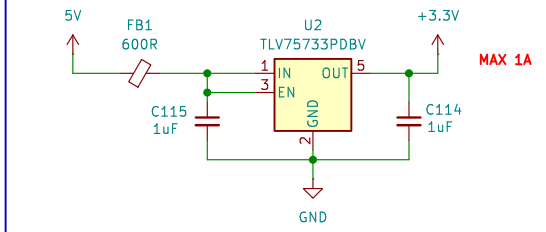
Title: CONTROLLER

Size: A4	Date: 2026-04-07	**Rev: 2**
KiCad E.D.A. 9.0.1		Id: 1/6

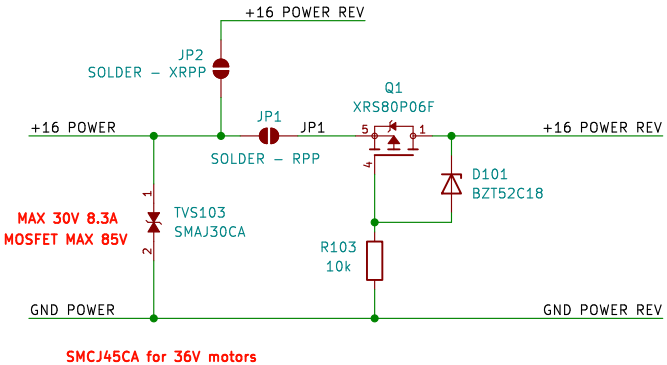
USBC



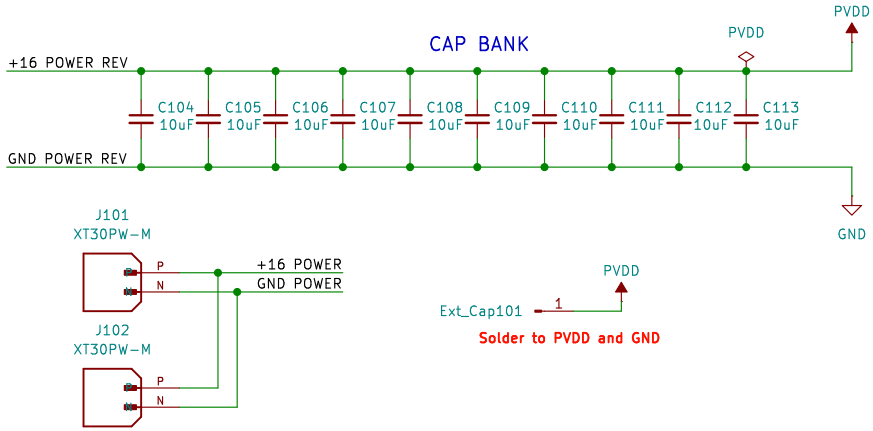
5V – 3V3



RPP



CAP BANK – BATT



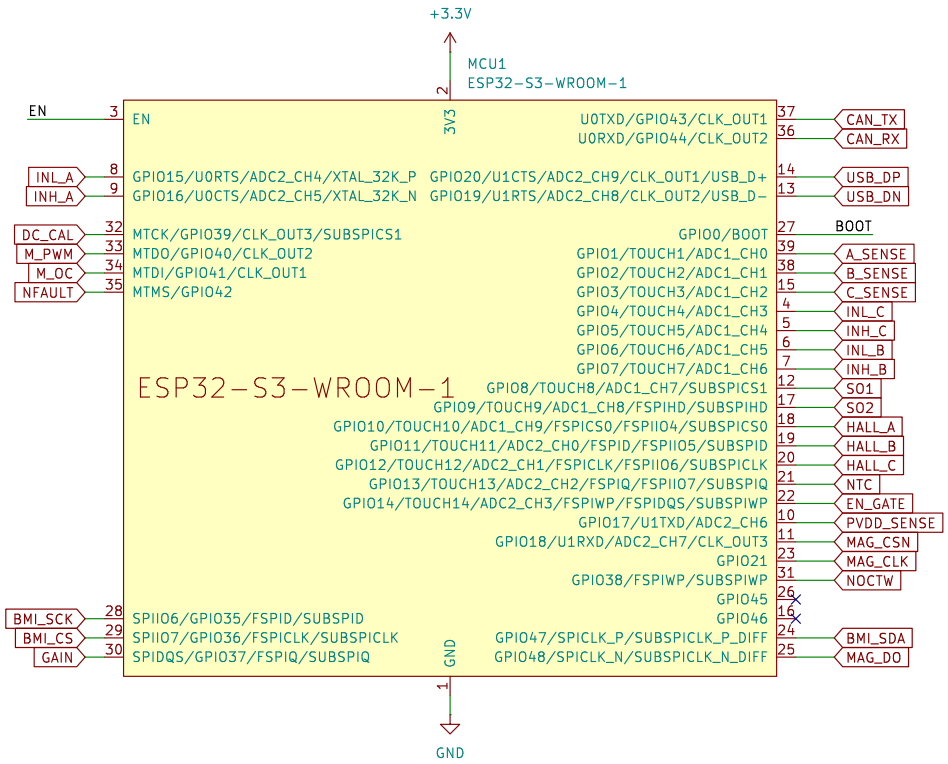
ANGEL PELEGRY VELASCO

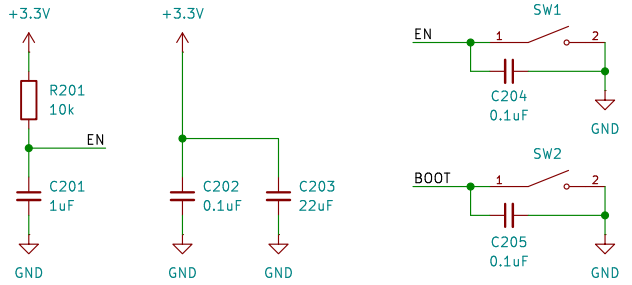
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File: POWER [1].kicad_sch

Title: POWER

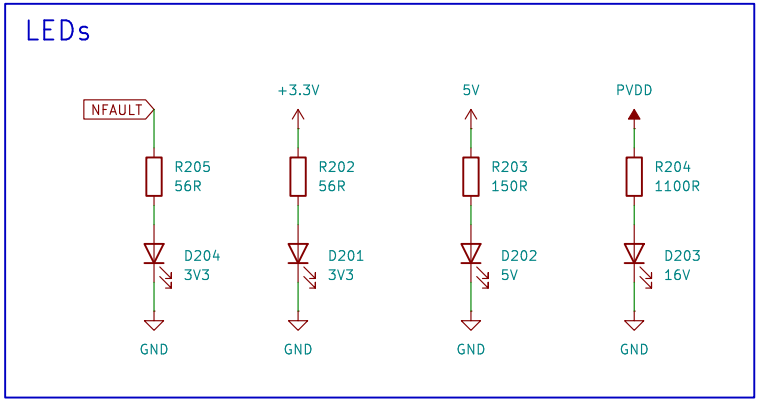
Size: A4	Date: 2026-04-07	**Rev: 2**
KiCad E.D.A. 9.0.1		Id: 1/6

ESP-32





LEDs





VSPI – LOW – 1.5V, HIGH 3.3V – GPIO 45

BOOT – LOW IF UART UPLOAD, HIGH BOOT FROM FLASH (Internal Pull up) – GPIO 0 – BOOT

STRAP – DEFAULT 0 (Internal pull down) NOT HIGH WHEN BOOT – GPIO 46

JTAG – DEFAULT 0 (USB – JTAG ENABLED) / HIGH FOR GPIO 3

INIT – SET C_SENSE – GPIO3 HIGH

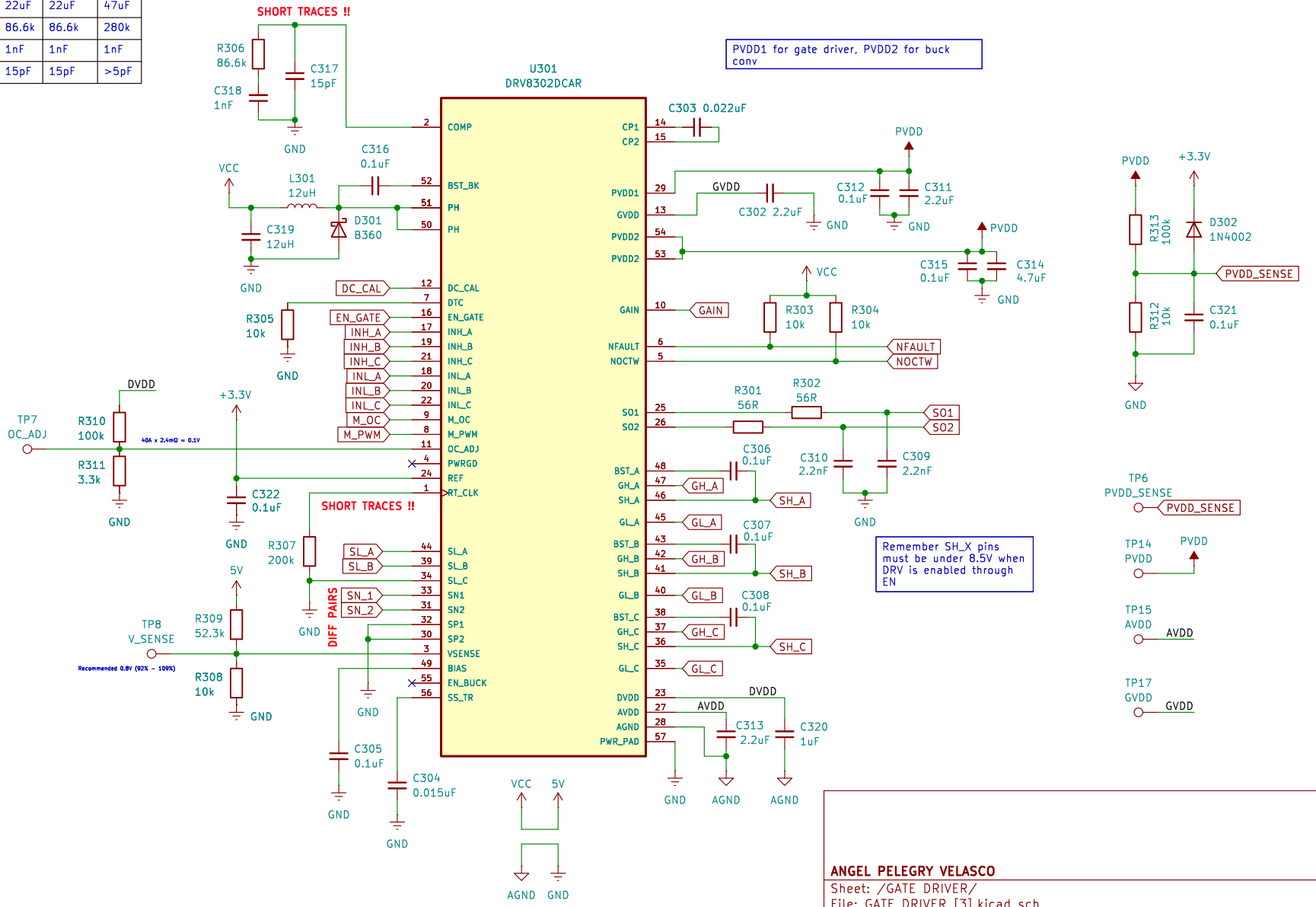
ANGEL PELEGRY VELASCO

Sheet: /CONTROLLER/
File: CONTROLLER [2].kicad_sch

Title: CONTROLLER

Size: A4	Date: 2026-04-07	**Rev: 2**
KiCad E.D.A. 9.0.1		Id: 2/6

Input Voltage	16	36	36
Output Voltage	5	5	16
L301 – Ind	12uH	22uH	42uH
Cout – C319	22uF	22uF	47uF
Rc – R306	86.6k	86.6k	280k
Cc – C318	1nF	1nF	1nF
Cf – C317	15pF	15pF	>5pF

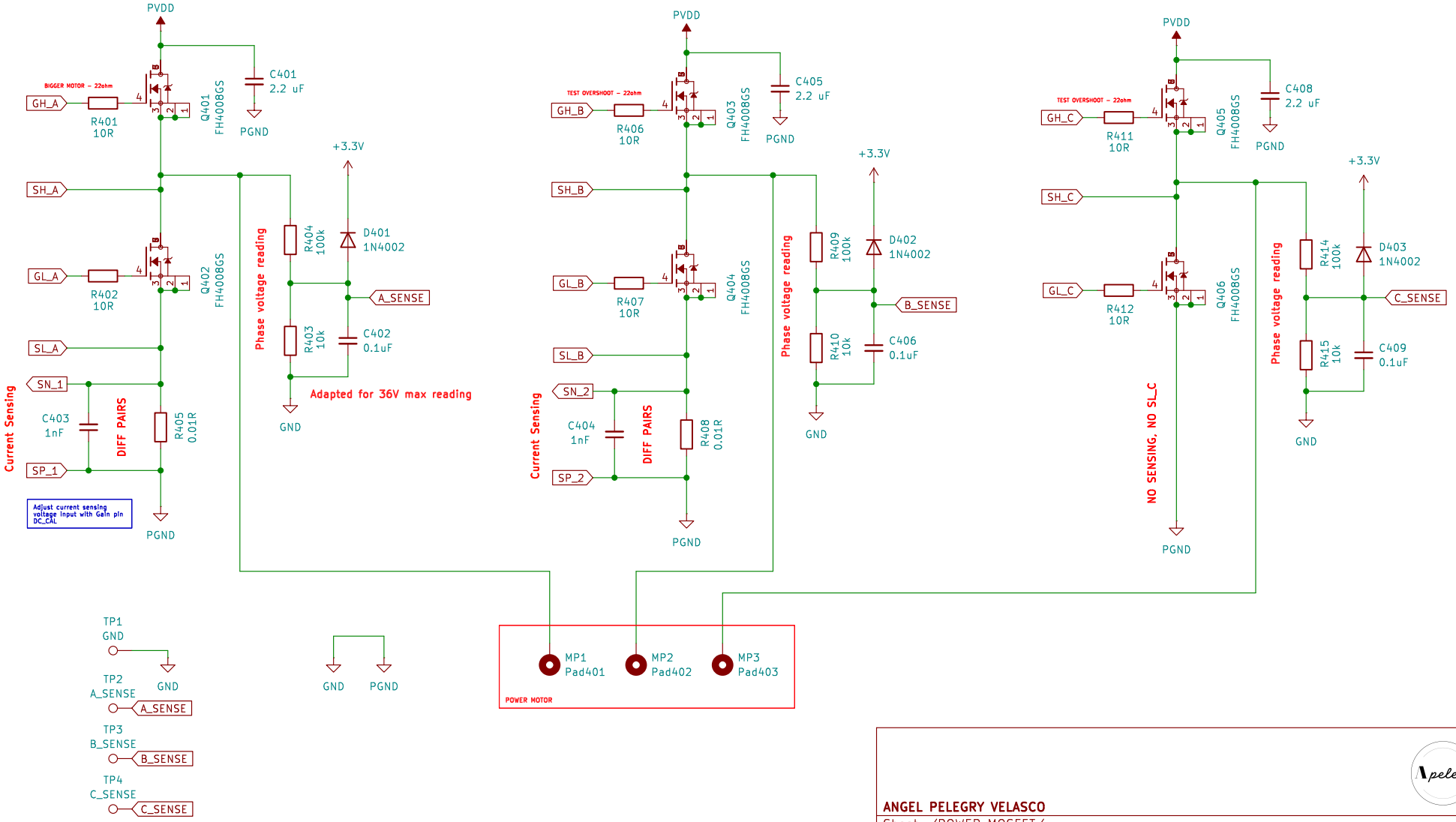


ANGEL PELEGRY VELASCO

Sheet: /GATE DRIVER/
File: GATE DRIVER [3].kicad_sch

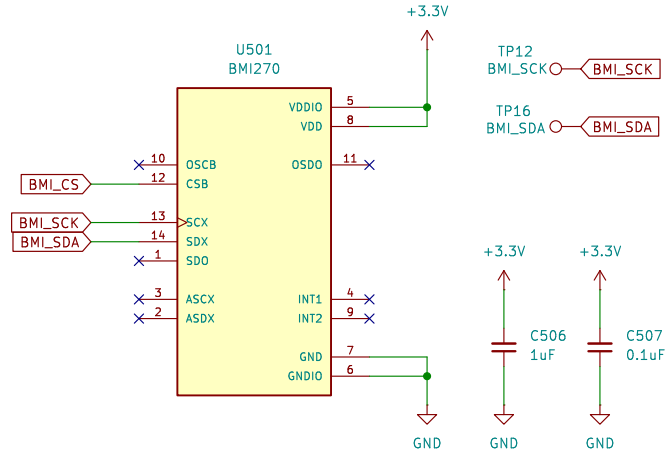
Title: GATE DRIVER

Size: A4	Date: 2026-04-07	**Rev: 2**
KiCad E.D.A. 9.0.1		Id: 3/6

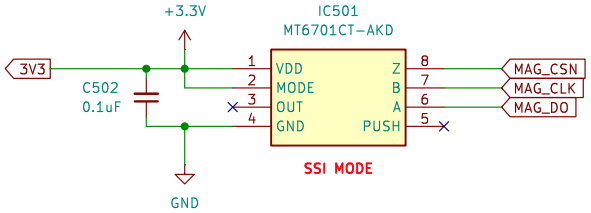


ANGEL PELEGRY VELASCO		
Sheet: /POWER MOSFET/		
File: POWER_MOSFET.kicad_sch		
Title: POWER MOSFET		
Size: A4	Date: 2026-04-07	**Rev: 2**
KiCad E.D.A. 9.0.1		Id: 4/6

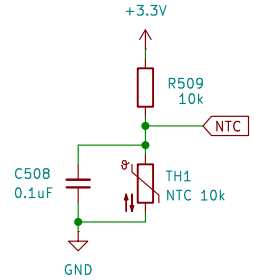
IMU + GYRO



MAGNETIC ENCODER

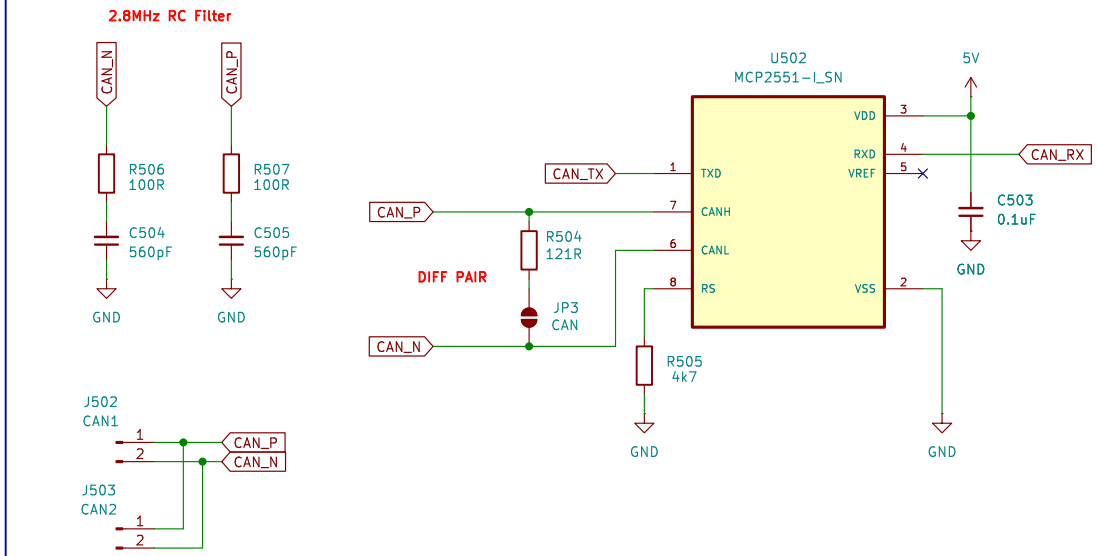


NTC THERMISTOR

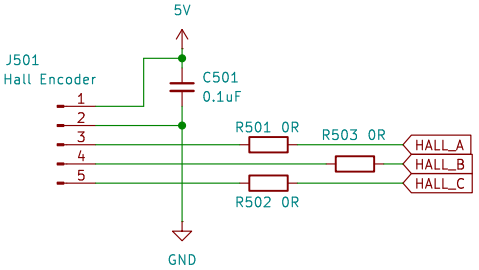


MP_1 MountingHole
MP_2 MountingHole
MP_3 MountingHole
MP_4 MountingHole

CAN TRANSCIEVER



HALL ENCODER



ANGEL PELEGRY VELASCO

Sheet: /SENSORS/
File: SENSORS.kicad_sch

Title: SENSORS

Size: A4	Date: 2026-04-07	**Rev: 2**
KiCad E.D.A. 9.0.1		Id: 5/6